

A complex function involving 10 Boolean variables can be defined as:

$$f(x_1, x_2, \dots, x_{10}) = (x_1 \oplus x_2) \wedge (x_3 \vee x_4) \oplus (x_5 \wedge x_6) \vee (x_7 \oplus x_8) \wedge (x_9 \oplus x_{10})$$

where:

- $\wedge$ represents the AND operation.
- $\vee$ represents the OR operation.
- $\oplus$ represents the XOR operation.

This function combines multiple logical operations to produce a complex decision boundary.

1. Construct a decision table listing all possible input combinations and their corresponding outputs for the given Boolean function.
2. Design an Artificial Neural Network (ANN) model with two hidden layers, each containing five nodes.
3. Develop a detailed mathematical representation of the ANN, including weight transformations, activation functions, and forward propagation equations.
4. Implement the ANN model from scratch in Python, incorporating weight initialization, forward propagation, backpropagation, and optimization techniques.
5. Performance evaluation (accuracy, confusion matrix, precision, recall, F1-score).
6. Convergence plot and performance metrics for different combinations of learning rate and number of epochs.
7. Apply GD, SGD, and ADAM optimizers and visualize their convergence rates through plots.
8. Experiment with different network architectures by varying the number of hidden layers (2, 3, and 4) and adjusting the number of neurons per layer (5, 10, and 15).